

EDUCATION

National Institute of Technology Patna

Dual Degree (B.Tech + M.Tech), Electrical Engineering

Courses: Operating Systems, Machine Learning, Deep Learning, Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Computer Vision, Embedded Systems

Patna, India

July 2023 - June 2028

TECHNICAL SKILLS

Programming Languages:: Python, C++, MATLAB, C, Java

Machine Learning:: PyTorch, TensorFlow, Scikit-Learn,CNNs, Vision Transformers (ViTs), Transfer Learning

Hardware-Aware ML:: Model Quantization, Model Pruning,hls4ml, FPGA Deployment, Vivado HLS

Robotics & Perception:: ROS 2, PX4, OpenCV,YOLOv8,NVIDIA Jetson, Sensor Fusion

Tools:: Git, Docker, Linux, CUDA, TensorBoard

EXPERIENCE

National Institute of Technology Patna

Machine Learning Researcher

Hardware-Aware Machine Learning: Conducting research on quantization, pruning, and FPGA deployment of deep neural networks using hls4ml.

Model Development: Developed CNN and Vision Transformer (ViT) models optimized for low-latency inference.

Deployment Optimization: Analyzed trade-offs between model accuracy and latency for efficient edge-AI deployment.

Patna, India

Jul 2025 - Present

Fast Machine Learning Lab (CERN Collaboration)

Open Source Contributor – hls4ml

Framework Development: Contributed to hls4ml for translating machine learning models for ultra-low-latency.

Research Collaboration: Collaborated with international researchers and maintainers through technical discussions, code reviews, and pull requests.

Remote

Jun 2026 - Present

Technology Innovation Hub (TiH), IIT Jodhpur

Project Intern

UAV Formation Control: Developed leader-follower UAV formation control algorithms.

Autonomous Navigation: Designed real-time perception and control pipelines for autonomous aerial navigation.

Research Dissemination: Presented research findings at the Advances in Robotics (AIR 2025) conference.

Jodhpur, India

May 2025 - Jul 2025

PROJECTS

Hardware-Aware Edge ML for Low-Latency FPGA Inference (GitHub):

Developed a hardware-aware optimization pipeline for Human Activity Recognition targeting FPGA deployment.

Applied fixed-point quantization and structured weight pruning of up to 80% while maintaining 93.5% classification accuracy.

Converted trained neural networks into FPGA-compatible firmware using hls4ml for low-latency inference.

KidneyViT: Vision Transformer for Kidney CT Classification (GitHub):

Developed a Vision Transformer (ViT) architecture for multi-class kidney CT image classification.

Achieved 99.8% validation accuracy using AdamW optimization and cosine learning-rate scheduling.

Generated attention-map visualizations to improve model interpretability and analyze model decisions.

Multi-Class Particle Classification using Machine Learning (GitHub):

Built supervised machine learning models for particle identification using 50,000 high-energy physics events.

Compared Random Forest and Neural Network architectures, achieving up to 96.85% classification accuracy.

Performed feature-importance analysis to identify dominant observables for particle discrimination.

PUBLICATIONS & PREPRINTS

Hardware-Efficient Squeeze-and-Excitation Networks for Anomaly Detection.

arXiv preprint, 2026 [in Process]

ACHIEVEMENTS

Solved 350+ algorithmic and data structure problems across LeetCode and other competitive programming platforms

Secured 2nd Prize among 100+ teams at NIT Patna Tech Fest for developing an agritech innovation solution.

Selected for the ISRO Robotics Challenge (IROC-U 2025) among 1100+ teams.

Awarded the Samsung Fellowship Award 2025 under the Samsung Innovation Award Program (ISWDP).